FUNCTION CODE :

```
function repf=spolarplot(G, omega)
f=omega/2/%pi;
repf=repfreq(G,f);
r=abs(repf);
theta=atan(imag(repf),real(repf));
polarplot(theta,r,style=2);
endfunction
```

MAIN PROGRAM :

```
clear;
clc;
xdel(winsid()); //close all windows
//please edit the path
//cd"/<your code directory>/"
//exec("spolarplot.sce")

cd"C:\Users\admin\Documents"
exec("spolarplot.sce")
s=%s;
omega=logspace(-1,3,1000);
G=syslin('c',10,0.05*s^3+0.6*s^2+s);
disp(G,'The given transfer function G(s)=');
spolarplot();
```

OUTPUT:

Scilab 6.0.2 Console

```
--> //polar plot of a linear system

--> //repf=spolarplot(G,omega)

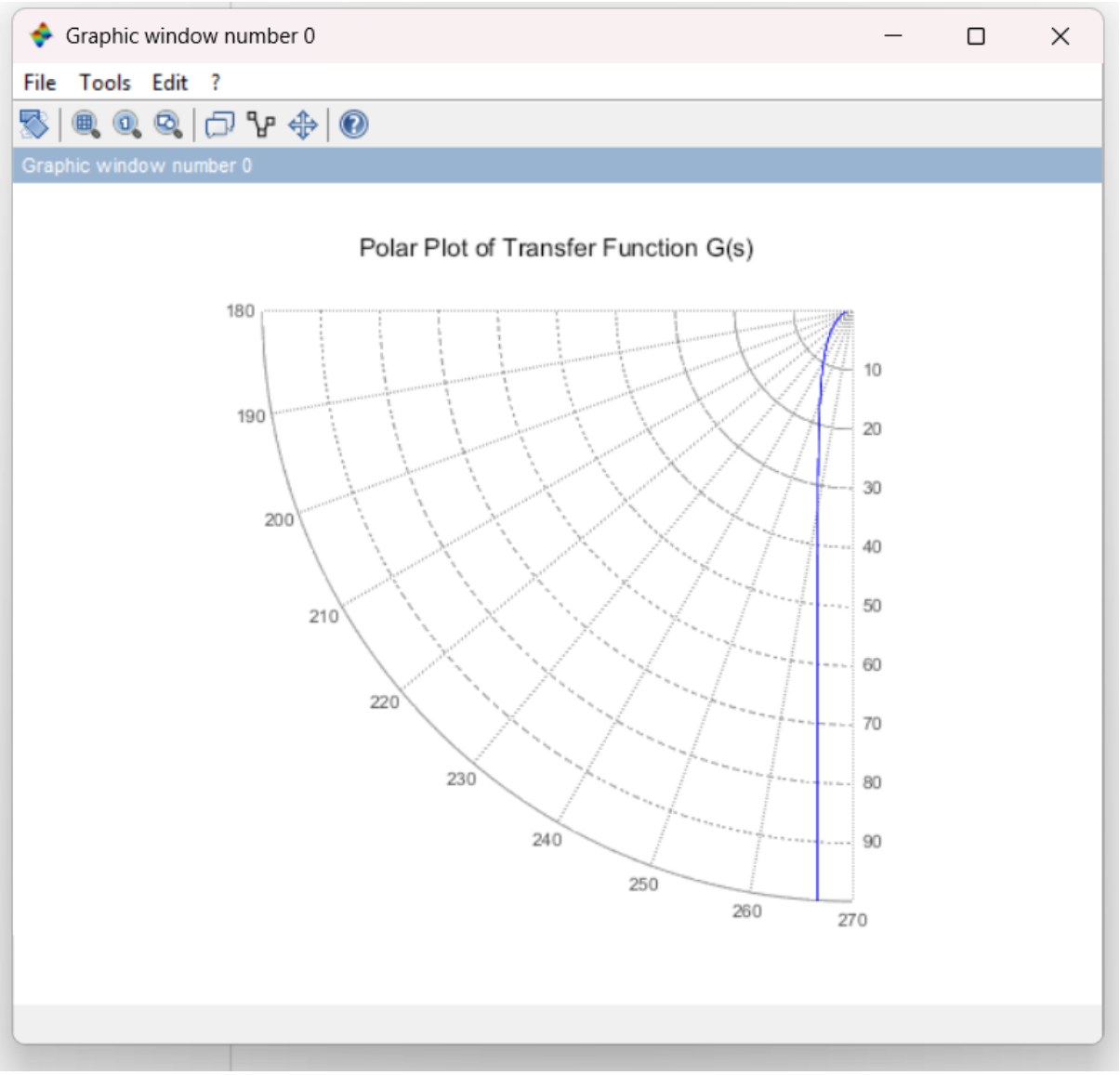
--> //G:linear system and omega: frquency in rad/s

--> //repf: complex frequency response

--> function repf=spolarplot(G,omega)
--> f=omega/2/%pi;
--> repf=repfreq(G,f);
--> r=abs(repf);
--> theta=atan(imag(repf),real(repf));
--> polarplot(theta,r,style=2);
--> endfunction

 The given transfer function is G(s)=


          10
   -----------------
            2       3
   s + 0.6s + 0.05s
```


Graphic window number 0
File Tools Edit ?
Graphic window number 0
Polar Plot of Transfer Function G(s)
180
190
200
210
220
230
240
250
260
270
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